

Integration

Week 5D

Overview

Integration by Substitution

- Key Ideas*
- *Integration by Simple Substitution*
 - *Simple Substitution with Definite Integrals*

Key Idea - Simple Substitution

There are times when a difficult integral can be made easier by using a substitution. Simple substitution is a technique for evaluating integrals by **reversing the chain rule**.

This technique involves making a substitution in order to simplify an integral. It has the effect of changing the variable and simplifying the integrand. We do this by letting a new variable, such as u (or t), represent a more complicated part of the expression we are trying to integrate. Choosing a good substitution is a skill that improves with practice.

Further, it is not simply a matter of changing the variable; **we must also take care to correctly substitute for the differential dx** .

Discussion Question:

$$\begin{aligned}\text{Differentiate } f(x) &= \frac{1}{5}(x^2 - 1)^5. \\ &= \cancel{\frac{1}{5}} \times \cancel{5} (x^2 - 1)^4 \times 2x \\ &= 2x(x^2 - 1)^4\end{aligned}$$

Write down the statement for integration.

$$\int 2x(x^2 - 1)^4 dx$$

What do we **recognise** between $\frac{d}{dx}\left(\frac{1}{5}(x^2 - 1)^5\right)$ and $\int 2x(x^2 - 1)^4 dx$?

Recall the Chain Rule $\frac{d}{dx}[f(g(x))] = f'(g(x)) \times g'(x)$ which was used to differentiate the composite function. $f(x) = \frac{1}{5}(x^2 - 1)^5$. If we carefully observe the answer, we obtain when we use the chain rule $f'(x) = 2x(x^2 - 1)^4$, we can learn to **recognise** when a function has this form, and so discover how to integrate such functions .

In the above example, $\int 2x(x^2 - 1)^4 dx$, we notice that

- that $f(g(x)) = (x^2 - 1)^4$

and

- and $g'(x) = 2x$

So we have a product of both $f(g(x))$ and $g'(x)$.

We use substitution when we have the product of

both $f(g(x))$ and $g'(x)$.

So, look for the product of

- a function inside another function i.e $f(g(x))$

AND

- the derivative of the inner function (which could be a constant) ie $g'(x)$.

The same method can be applied to other composite functions that we will study in this subject.

Given an integral of the form $\int f(g(x))g'(x) dx$

Step 1: Let $u = g(x)$

Step 2: Find $\frac{du}{dx} = g'(x)$, so $du = g'(x) dx$.

[You may have to adjust constant factors]

Step 3: Make the substitutions $u = g(x)$ and $du = g'(x) dx$ in the original integral, to get $\int f(u) du$.

The integral must now be entirely in terms of u , including the ' dx '.

Step 4: Evaluate the new integral (with respect to u) $\int f(u) du$.

Step 5: In the answer back substitute each occurrence of variable u by $g(x)$.
The answer must now be entirely in terms of x , or the original variable.

Example: Integrate $\int 2x(x^2 - 1)^4 dx$

In this example we make the substitution $u = x^2 - 1$, in order to simplify the term that is raised to the power of 4.

Step 1: Let $u = x^2 - 1$

We shall see that the rest of the integrand, $2x dx$, will be taken care of automatically in the substitution process, and that this is because $2x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = x^2 - 1$

Step 2: $\frac{du}{dx} = 2x$ and $du = 2x dx$

So, substitute u for $x^2 - 1$, and with $2x dx = du$ in $\int 2x(x^2 - 1)^4 dx$

$$\begin{aligned} \text{Step 3: } \int 2x(x^2 - 1)^4 dx &= \int \underbrace{(x^2 - 1)^4}_{u^4} \underbrace{2x dx}_{du} \\ &= \int u^4 du \end{aligned}$$

This expression clearly looks much simpler function to integrate than $(x^2 - 1)^4$.

Now integrate the new function with respect to u .

$$\text{Step 4: } \int u^4 du = \frac{1}{5}u^5 + C$$

We now back substitute each occurrence of variable u by $x^2 - 1$, giving

$$\text{Step 5: } \frac{1}{5}(x^2 - 1)^5 + C$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{1}{5}(x^2 - 1)^5 \right)$

Example: Integrate $\int 3x^2 \sqrt{1+x^3} \, dx$

In this example we make the substitution $u = 1 + x^3$, in order to simplify the square root term.

Step 1: Let $u = 1 + x^3$

We shall see that the rest of the integrand, $3x^2 \, dx$, will be taken care of automatically in the substitution process, and that this is because $3x^2$ is the derivative of that part of the integrand used in the substitution, i.e. $u = 1 + x^3$

Step 2: $\frac{du}{dx} = 3x^2$ and $du = 3x^2 \, dx$

So, substitute u for $1 + x^3$, and with $3x^2 \, dx = du$ in $\int 3x^2 \sqrt{1+x^3} \, dx$

$$\begin{aligned} \text{Step 3: } \int 3x^2 \sqrt{1+x^3} \, dx &= \int \underbrace{\sqrt{1+x^3}}_{\sqrt{u}} 3x^2 \, dx \\ &= \int \sqrt{u} \, du \end{aligned}$$

This expression clearly looks much simpler function to integrate than $\sqrt{1+x^3}$.

Now integrate the new function with respect to u .

$$\begin{aligned}\text{Step 4: } \int \sqrt{u} \, du &= \int u^{1/2} \, du \\ &= \frac{u^{3/2}}{3/2} + C \\ &= \frac{2u^{3/2}}{3} + C\end{aligned}$$

We now back substitute each occurrence of variable u by $1 + x^3$, giving

$$\text{Step 5: } \frac{2(1+x^3)^{3/2}}{3} + C \text{ or } \frac{2}{3}(1+x^3)^{3/2} + C$$

Note: You can check this solution by differentiation $\frac{d}{dx} \left(\frac{1}{5}(x^2 - 1)^5 \right)$

Example: Integrate $\int x(x^2 - 3)^5 dx$

In this example we make the substitution $u = (x^2 - 3)$, in order to simplify the square root term.

Step 1: Let $u = (x^2 - 3)$

We shall see that the rest of the integrand, $x dx$, will be taken care of automatically in the substitution process, and that this is because $x dx$ is the derivative of that part of the integrand used in the substitution, i.e. $u = (x^2 - 3)$. There is a slight complication, however. **The integral does not have the constant factor of 2, so we need to adjust it. TED??**

Step 2: $\frac{du}{dx} = 2x$ and $\frac{du}{2} = x dx$

So, substitute u for $(x^2 - 3)$, and with $\frac{du}{2} = x dx$ in $\int x(x^2 - 3)^5 dx$

$$\begin{aligned}\text{Step 3: } \int x(x^2 - 3)^5 dx &= \int \overbrace{(x^2 - 3)^5}^u x dx \\ &= \int u^5 \frac{du}{2} \\ &= \frac{1}{2} \int u^5 du\end{aligned}$$

This expression clearly looks much simpler function to integrate than $(x^2 - 3)^5$.

Now integrate the new function with respect to u .

$$\begin{aligned}\text{Step 4: } \frac{1}{2} \int u^5 du &= \frac{1}{2} \times \frac{u^6}{6} + C \\ &= \frac{u^6}{12} + C\end{aligned}$$

We now back substitute each occurrence of variable u by $(x^2 - 3)$, giving

$$\text{Step 5: } \frac{(x^2 - 3)^6}{12} + c \text{ or } \frac{1}{12} (x^2 - 3)^6 + c$$

Note: You can check this solution by differentiation $\frac{d}{dx} \left(\frac{1}{12} (x^2 - 3)^6 \right)$

Exercise:

Find the indefinite integrals $\int 3x^2 \sqrt{4+x^3} dx$.

1. let $u = 4+x^3$

2. $\frac{du}{dx} = 3x^2 \Rightarrow du = 3x^2 dx$

3. $\int 3x^2 \sqrt{4+x^3} dx = \int \sqrt{u} du$

4. $\int \sqrt{u} du = \int u^{\frac{1}{2}} du$
 $= \frac{2u^{\frac{3}{2}}}{\frac{3}{2}} + C$

5. $\frac{2(4+x^3)^{\frac{3}{2}}}{\frac{3}{2}} + C$ or $\frac{2\sqrt{(4+x^3)^3}}{3} + C$

Exercise:

Find the indefinite integrals $\int \frac{x+1}{(x^2+2x)^3} dx = \int (x+1)(x^2+2x)^{-3} dx$

$$\text{let } u = x^2 + 2x$$

$$\begin{aligned} \frac{du}{dx} &= 2x+2 \Rightarrow du = (2x+2) dx \\ du &= 2(x+1) dx \\ \frac{du}{2} &= (x+1) dx \end{aligned}$$

$$\begin{aligned} \int \frac{x+1}{(x^2+2x)^3} dx &= \int \frac{1}{u^3} \frac{du}{2} \\ &= \frac{1}{2} \int \frac{1}{u^3} du \\ &= \frac{1}{2} \int u^{-3} du \\ &= \frac{1}{2} \frac{u^{-2}}{-2} + C \\ &= -\frac{1}{4u^2} + C = -\frac{1}{4(x^2+2x)^2} + C \end{aligned}$$

Key Idea - Simple Substitution with Definite Integrals

When using the **substitution method with definite integrals**, it is important to handle the limits of integration carefully.

When a substitution is made, the limits must also be adjusted accordingly. There are two options:

I recommend the first!

1. Change the limits from x -values to u -values, rewriting the integral entirely in terms of u .

This is the recommended method.

2. Evaluate the integral first as an indefinite integral, this answer is in terms of x and then substitute back in the original x -limits.

Example: Evaluate the definite integral $\int_0^1 3x^2(x^3+1)^4 dx$.

$$\text{Let } u = x^3 + 1, \quad \frac{du}{dx} = 3x^2$$

$$\text{Limits: when } x = 0, u = 0^3 + 1 = 1$$

$$\text{when } x = 1, u = 1^3 + 1 = 2$$

$$\text{Step 3: } \int_{x=0}^{x=1} 3x^2(x^3+1)^4 dx = \int_{u=1}^{u=2} u^4 du$$

$$= \left[\frac{u^5}{5} \right]_1^2$$

$$= \left(\frac{2^5}{5} - \frac{1^5}{5} \right)$$

$$= \left(\frac{32}{5} - \frac{1}{5} \right) = \frac{31}{5}$$

Note: That the original limits of integration are limits on the variable x , not u . To emphasise this, they have been written explicitly as $x = 0$ and $x = 1$. When we integrate with respect to the variable u , the limits must be written in terms of u too.

Note: You do not need to write the variable x or u for the limits, only the actual limits

Example: Evaluate the definite integral $\int_0^1 x^2 \sqrt{x^3 + 4} \, dx$.

Let $u = x^3 + 4$

$$\frac{du}{dx} = 3x^2 \quad \text{and} \quad \frac{du}{3} = x^2 dx$$

Limits: when $x = 0$, $u = 0^3 + 4 = 4$

when $x = 1$, $u = 1^3 + 4 = 5$

$$\begin{aligned} \int_0^1 x^2 \sqrt{x^3 + 4} \, dx &= \int_4^5 x^2 \sqrt{x^3 + 4} \, dx \\ &= \int_4^5 \sqrt{u} \, \frac{du}{3} = \frac{1}{3} \int_4^5 u^{1/2} \, du \\ &= \frac{1}{3} \left[\frac{u^{3/2}}{3/2} \right]_4^5 = \frac{1}{3} \left[\frac{2}{3} u^{3/2} \right]_4^5 \\ &= \frac{2}{9} \left[u^{3/2} \right]_4^5 = \frac{2}{9} (5^{3/2} - 4^{3/2}) \end{aligned}$$

Exercise:

Evaluate the definite integral $\int_1^2 (3x^2 - 3)(x^3 - 3x + 2)^{\frac{5}{2}} dx$

Limits: $x=1$ $u = 1^3 - 3 \cdot 1 + 2$
 $= 1 - 3 + 2$
 $= 0$

$x=2$ $u = 2^3 - 3 \cdot 2 + 2$
 $= 8 - 3 + 2$
 $= 4$

Let $u = x^3 - 3x + 2$

$\frac{du}{dx} = 3x^2 - 3 \Rightarrow du = (3x^2 - 3)dx$

$\therefore \int_0^4 u^{\frac{5}{2}} du = \left[\frac{2u^{\frac{7}{2}}}{7} \right]_0^4 = \frac{2}{7} \left[u^{\frac{7}{2}} \right]_0^4$

$= \frac{2}{7} \left(4^{\frac{7}{2}} - 0^{\frac{7}{2}} \right)$

$= \frac{2}{7} \times 2^7$

$= \frac{256}{7}$

Exercise:

Evaluate the definite integral $\int_0^2 x\sqrt{x^2+1} \, dx$.

$$\text{Limits: } x=0 \quad u=0^2+1=1$$

$$x=2 \quad u=2^2+1=5$$

$$\text{let } u = x^2 + 1$$

$$\frac{du}{dx} = 2x \Rightarrow du = 2x \, dx$$

$$\Rightarrow \frac{du}{2} = x \, dx$$

$$\therefore \int_0^2 x\sqrt{x^2+1} \, dx = \int_1^5 \sqrt{u} \frac{du}{2} = \frac{1}{2} \int_1^5 \sqrt{u} \, du$$

$$= \frac{1}{2} \int_1^5 u^{1/2} \, du$$

$$= \frac{1}{2} \left[\frac{2u^{3/2}}{3} \right]_1^5 = \frac{1}{2} \times \frac{2}{3} (u^{3/2}) \Big|_1^5$$

$$= \frac{1}{3} (u^{3/2}) \Big|_1^5$$

$$= \frac{1}{3} (5^{3/2} - 1^{3/2}) = \frac{1}{3} (\sqrt{5^3} - 1)$$

$$= \frac{1}{3} (\sqrt{125} - 1)$$